

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CL)

April 2013

CL 208 Advance Surveying

Date: 23.04.2013, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) Derive distance and elevation formula for incline line of sight, staff held vertical. [04]
(b) Define. (i) Mistakes ii) True value iii) Most probable value [03]
- Q - 2 (a) A tacheometer is setup at point A fitted with an anallatic lens with an instrument constant of 100, If staff was held vertical determine the gradient from point P to Q [05]

Inst. Station	Staff Station	Bearing	Vertical angle	Hair readings
A	P	345°	+15°	0.915, 1.750, 2.585
	Q	75°	+10°	0.760, 2.240, 3.715

- (b) Determine the most probable value of angles of a triangle PQR using the method of correlates [05]
 $\angle P = 58^\circ 24' 36''$ Weight 3 $\angle Q = 52^\circ 12' 43''$ Weight 3
 $\angle R = 69^\circ 22' 45''$ Weight 2
- (c) Enlist the application of GIS and explain any two in detail. [04]

OR

- Q - 2 (a) Derive elevation and distance formula for tangential method, when one angle is elevation and other angle in depression. Also write equation to find R.L of staff station. [05]
(b) Find the most probable value of the angles A, B, C of a triangle ABC from the following observation by forming normal equation. [05]
 $\angle A = 77^\circ 14' 20''$ Weight 4 $\angle B = 49^\circ 40' 35''$ Weight 3
 $\angle C = 53^\circ 04' 52''$ Weight 4
(c) Explain the working of GPS. [04]
- Q - 3 (a) Derive distance and elevation formula for stadia method when line of sight is incline θ is +ve and staff held normal [04]
(b) Explain theory of least squares. [04]
(c) Define remote sensing and give difference between active and passive system of remote sensing. [06]

OR

- Q - 3 (a) Give the difference between tangential method and Stadia method. [04]

(b) Write any four laws of weight. [04]

(c) Explain energy interaction in atmosphere and earth surface. [06]

SECTION – II

Q - 4 (a) Define following [05]

(i) Azimuth (ii) Fiducial Marks (iii) Ground nadir (iv) Signals (v) Station marks

(b) Find the local mean time at a place having longitude of $94^{\circ} 20'E$, When the standard time is $10^h 06^m 18^s$ and the standard meridian is $78^{\circ} 30'E$. [02]

Q - 5 (a) Determine level of high object, when Base of the object inaccessible, Instrument stations in the same vertical plane as the elevated object. When B is lower than that of A and A is near to the object. [07]

(b) Explain scale of vertical and tilted photographs. [07]

OR

Q - 5 (a) The following observation was made on top of tower located on hill to compute elevation of hill. The height of the tower is 10m. Calculate elevation of hill. The instrument stations were 100m apart. [07]

Inst. St	Staff reading on B.M	Vertical angle on top of tower	R.L of Bench Mark
P	3.50	25°	320m
Q	2.25	15°	

(b) Define Parallax and derive parallax equation. [07]

Q - 6 (a) Explain principal of triangulation. Give the criteria for selection triangulation station. [07]

(b) Draw the celestial sphere and show following position [07]

(i) Earth centre, Zenith, Nadir, (ii) Azimuth of celestial body (iv) Altitude of celestial body (v) Declination of celestial body (vi) Hour circle, Vertical circle (vii) Observer meridian, North and south celestial pole.

OR

Q - 6 (a) What is base line explain procedure to extend the base line. [07]

(b) Determine the declination and hour angle of stars for the following. [07]

Altitude of star = 22° , Azimuth A = $150^{\circ} E$, Latitude of observer = $49^{\circ} 20'N$
